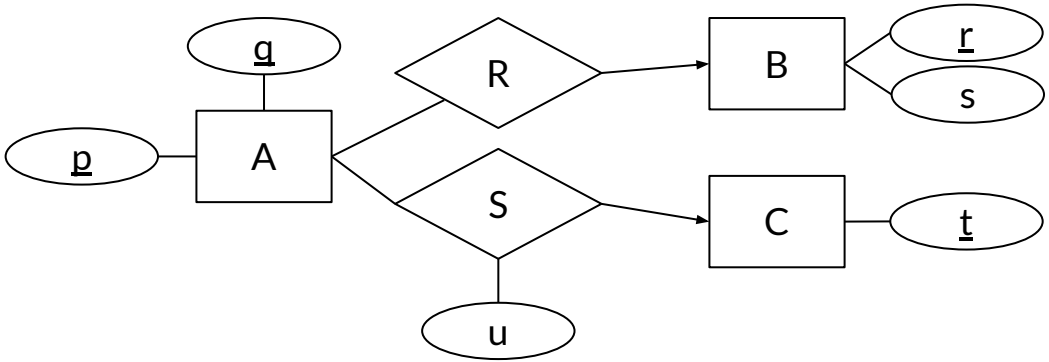
CSC 370: Databases Projecting FD's

Sean Chester • University of Victoria

Which FDs are implied?



- ❑ $p \rightarrow u$
- ❑ $pq \rightarrow s$
- ❑ $r \rightarrow s$
- ❑ $A \rightarrow BC$
- ❑ $t \rightarrow u$
- ❑ $u \rightarrow u$

Allotted Time: 60 seconds

Today's Goal:

You can correctly project a set of functional dependencies onto a sub-relation and produce a minimal basis thereof

Outline

1. Review closures of attribute sets
2. Introduce closures of FD sets
3. Describe rules for a minimal basis
4. Project a set of FD's onto a sub-relation
5. Summary / Exercise

Previously...

Review: Functional Dependency

Consider a schema $R(C_1, C_2, ..., C_k)$ and let $C = \{C_1, C_2, ..., C_k\}$ be the set of attributes

Let $A = \{A_1, A_2, ..., A_n\} \subseteq C$ and $B = \{B_1, B_2, ..., B_m\} \subseteq C$ be subsets of R's attributes.

We say that A *functionally determines B*, denoted $A \rightarrow B$, if and only if:

> For every two possible tuples of R,
> if they have the same values for *all*
> attributes in A, then they must have
> the same values for all attributes in B.

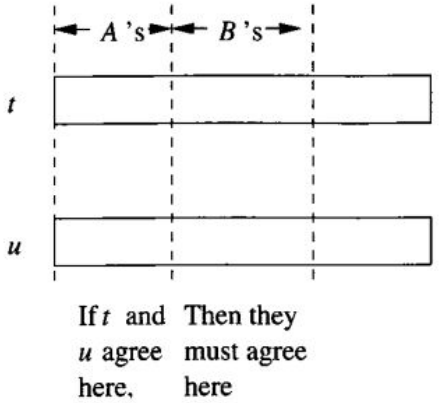


Figure 3.1: The effect of a functional dependency on two tuples.

Source: Garcia-Molina et al. (2007), §3.1.1

Some rules that we can apply to FD's

Because functional dependencies are logical inferences, we can apply a number of logic rules to expand our set of FD's

Triviality
We can always assert a trivial FD of the form $A \rightarrow B$ if $B \subseteq A$

For example:
writer song_name → writer

Semi-triviality
Given an FD of the form $A \rightarrow B$ and C = B∩A, we can drop C from B

For example:
album song_name → album run_length

Can be simplified to:
album song_name → run_length

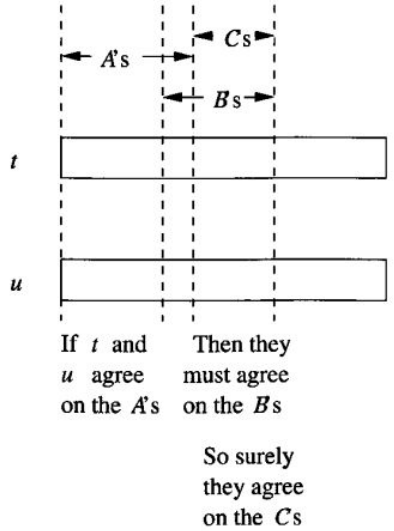


Figure 3.3: The trivial-dependency rule

Source: Garcia-Molina et al. (2007), §3.2.3

The *closure* of an attribute set

Using these rules, a natural question that we might ask is:

Given a set of attributes X, what is the maximum set of attributes that we can functionally determine from them?

This is called the closure of X, and is denoted X^+

Consider that you have the following FD's:
A→C
BC→D
C→*B*

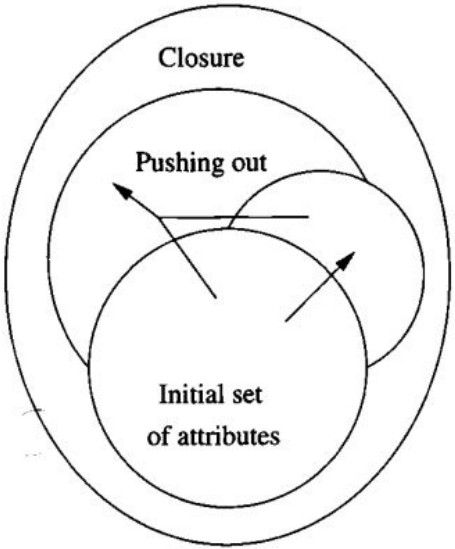


Figure 3.4: Computing the closure of a set of attributes

What is the closure of {A}, i.e., what is $\{A\}^+$?

1. Observe that A→C; so, $\{A\}^+ \supseteq \{A,C\}$
2. Observe that C→B; so, $\{A\}^+ \supseteq \{A,B,C\}$
3. Observe that BC→D; so, $\{A\}^+ \supseteq \{A,B,C,D\}$
4. At this point, we cannot exploit any more rules; so, we conclude that $\{A\}^+ = \{A,B,C,D\}$

Source: Garcia-Molina et al. (2007), §3.2.4

Recall: Attribute Closures (example 2)

In *relational* design, we specify (among perhaps other things):

- A relation schema, i.e., name and set of attributes
- A set of functional dependencies, i.e., constraints that capture dependencies between attributes

Given these, we can calculate the *closure* of any set of attributes using our rules of *transitivity* and (semi-)*triviality*. For example:

R(A,B,C,D,E)
A → B
B → D
CD → E

What is the closure of {AC}, i.e., $\{AC\}^+$?

$\{AC\}^+ \supseteq \{A,C\}$ // from triviality
$\{AC\}^+ \supseteq \{A,B,C\}$ // b/c A → B
$\{AC\}^+ \supseteq \{A,B,C,D\}$ // b/c B → D
$\{AC\}^+ \supseteq \{A,B,C,D,E\}$ // b/c CD → E

Therefore, {AC} is a *superkey*. (It also happens to be a *key*.)

Source: Garcia-Molina et al. (2007), §3.1

Bases

Closing Set of a set of FD's

The *closing set* of a *set of functional dependencies* (as opposed to a set of attributes) is:

- The closure of all possible subsets of the attributes

For example, given:
R(A,B,C)
$A \rightarrow B$
$C \rightarrow B$

The closing set for $\{A \rightarrow B, C \rightarrow B\}$ is:

Source: Garcia-Molina et al. (2007), §3.2.7

Closing Set of a set of FD's

The *closing set* of a *set of functional dependencies* (as opposed to a set of attributes) is:

- The closure of all possible subsets of the attributes

For example, given:
R(A,B,C)
$A \rightarrow B$
$C \rightarrow B$

The closing set for $\{A \rightarrow B, C \rightarrow B\}$ is:

$\{\}^+ = \{\}$
$\{A\}^+ = \{A,B\}$
$\{B\}^+ = \{B\}$
$\{C\}^+ = \{B,C\}$
$\{A,B\}^+ = \{A,B\}$
$\{A,C\}^+ = \{A,B,C\}$
$\{B,C\}^+ = \{B,C\}$
$\{A,B,C\}^+ = \{A,B,C\}$

Source: Garcia-Molina et al. (2007), §3.2.7

Basis of a set of FD's

We can then say that two *sets of functional dependencies* are equivalent iff:

- They have the same *closing set*
- When two sets of FD's are equivalent, we say they are a *basis* for each other

Given:
R(A,B,C)
$A \rightarrow B$
$C \rightarrow B$

What other sets of FD's form a basis
for this set ($\{A \rightarrow B, C \rightarrow B\}$)

Source: Garcia-Molina et al. (2007), §3.2.7

Basis of a set of FD's

We can then say that two *sets of functional dependencies* are equivalent iff:

- They have the same *closing set*
- When two sets of FD's are equivalent, we say they are a *basis* for each other

Given:
R(A,B,C)
$A \rightarrow B$
$C \rightarrow B$

What other sets of FD's form a basis for this set ($\{A \rightarrow B, C \rightarrow B\}$)

Some examples:

$A \rightarrow AB$
$C \rightarrow BC$

$A \rightarrow B$
$B \rightarrow B$
$C \rightarrow B$

$AB \rightarrow B$
$A \rightarrow AB$
$C \rightarrow B$

Source: Garcia-Molina et al. (2007), §3.2.7

Basis of a set of FD's

We can then say that two *sets of functional dependencies* are equivalent iff:

- They have the same *closing set*
- When two sets of FD's are equivalent, we say they are a *basis* for each other

Given:
R(A,B,C)
$A \rightarrow B$
$C \rightarrow B$

What other sets of FD's form a basis for this set ($\{A \rightarrow B, C \rightarrow B\}$)

Some examples:

$A \rightarrow AB$	$A \rightarrow B$	$AB \rightarrow B$
$C \rightarrow BC$	$B \rightarrow B$	$A \rightarrow AB$
	$C \rightarrow B$	$C \rightarrow B$

If it's clear that two sets of FD's can be equivalent, then we can ask if any of them are better than any others.

Source: Garcia-Molina et al. (2007), §3.2.7

Basis of a set of FD's: Example 2

Given two sets of FD's, F and F', we say that F' is a *basis* for F (and vice versa) iff:

The closure in every subset of attributes is identical in F and F'

We also say that F and F' are *equivalent*

Let's see some more examples:

F	**F'**
$A \rightarrow B$	$A \rightarrow D$
$B \rightarrow D$	$D \rightarrow B$
$D \rightarrow A$	$B \rightarrow A$

Can we confirm these are equivalent?

Source: Garcia-Molina et al. (2007), §3.2.7

These sets of FDs are equivalent.

Set F

$A \rightarrow BCD$

Set F'

$A \rightarrow B$
$A \rightarrow C$
$A \rightarrow D$

❏ true ❏ false

Allotted Time: 60 seconds

These sets of FDs are equivalent.

Set F

$A \rightarrow BC$
$B \rightarrow AD$

Set F'

$A \rightarrow B$
$B \rightarrow ACD$

❏ true ❏ false

Allotted Time: 90 seconds

Minimal bases

Minimal Basis

Given a set of FD's *F*, we say that they form *a minimal basis* iff:

- *Every FD has a singleton right-hand side*
- *If any FD is removed, we no longer have a basis*
- *If we remove any attribute from the left-hand side of any FD, we no longer have a basis*

In other words, there is no simpler set of bases that form a basis for *F*.

Source: Garcia-Molina et al. (2007), §3.2.7

Minimal Basis

Given a set of FD's *F*, we say that they form *a minimal basis* iff:

- *Every FD has a singleton right-hand side*
- *If any FD is removed, we no longer have a basis*
- *If we remove any attribute from the left-hand side of any FD, we no longer have a basis*

In other words, there is no simpler set of bases that form a basis for *F*.

Are both of these minimal bases?

F	**F'**
$A \rightarrow B$	$A \rightarrow D$
$B \rightarrow D$	$D \rightarrow B$
$D \rightarrow A$	$B \rightarrow A$

Source: Garcia-Molina et al. (2007), §3.2.7

Minimal Basis: Example 2

Given a set of FD's *F*, we say that they form *a minimal basis* iff:

- *Every FD has a singleton right-hand side*
- *If any FD is removed, we no longer have a basis*
- *If we remove any attribute from the left-hand side of any FD, we no longer have a basis*

In other words, there is no simpler set of bases that form a basis for *F*.

Are both of these minimal bases?

F	**F'**	**F"**
$A \rightarrow BD$	$AB \rightarrow D$	$A \rightarrow D$
$B \rightarrow DA$	$D \rightarrow B$	$D \rightarrow B$
$D \rightarrow AB$	$B \rightarrow A$	$B \rightarrow A$
		$D \rightarrow A$

Source: Garcia-Molina et al. (2007), §3.2.7

Calculating a Minimal Basis

To calculate a minimal basis, we apply the following steps:

1. First apply the splitting rule to meet the first condition
2. Then try to remove each FD and see if we still have a basis
3. Finally try to remove attributes from the left-hand side of the remaining FD's

Source: Garcia-Molina et al. (2007), §3.1.1

Calculating a Minimal Basis

To calculate a minimal basis, we apply the following steps:

1. First apply the splitting rule to meet the first condition
2. Then try to remove each FD and see if we still have a basis
3. Finally try to remove attributes from the left-hand side of the remaining FD's

F
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow AC$
$C \rightarrow A$
$A \rightarrow C$

Source: Garcia-Molina et al. (2007), §3.1.1

Calculating a Minimal Basis

To calculate a minimal basis, we apply the following steps:

1. First apply the splitting rule to meet the first condition
2. Then try to remove each FD and see if we still have a basis
3. Finally try to remove attributes from the left-hand side of the remaining FD's

F

$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow AC$
$C \rightarrow A$
$A \rightarrow C$

F'

$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow A$
$D \rightarrow C$
$C \rightarrow A$
$A \rightarrow C$

Source: Garcia-Molina et al. (2007), §3.1.1

Calculating a Minimal Basis

To calculate a minimal basis, we apply the following steps:

1. First apply the splitting rule to meet the first condition
2. Then try to remove each FD and see if we still have a basis
3. Finally try to remove attributes from the left-hand side of the remaining FD's

F
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow AC$
$C \rightarrow A$
$A \rightarrow C$

F'
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow A$
$D \rightarrow C$
$C \rightarrow A$
$A \rightarrow C$

F"
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow A$
~~$D \rightarrow C$~~
$C \rightarrow A$
$A \rightarrow C$

Source: Garcia-Molina et al. (2007), §3.1.1

Calculating a Minimal Basis

To calculate a minimal basis, we apply the following steps:

1. First apply the splitting rule to meet the first condition
2. Then try to remove each FD and see if we still have a basis
3. Finally try to remove attributes from the left-hand side of the remaining FD's

F
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow AC$
$C \rightarrow A$
$A \rightarrow C$

F'
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow A$
$D \rightarrow C$
$C \rightarrow A$
$A \rightarrow C$

F''
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow A$
~~$D \rightarrow C$~~
$C \rightarrow A$
$A \rightarrow C$

F'''
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow A$
$D \rightarrow C$
$C \rightarrow A$
$A \rightarrow C$

Source: Garcia-Molina et al. (2007), §3.1.1

Calculating a Minimal Basis

To calculate a minimal basis, we apply the following steps:

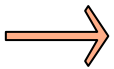
1. First apply the splitting rule to meet the first condition
2. Then try to remove each FD and see if we still have a basis
3. Finally try to remove attributes from the left-hand side of the remaining FD's

F
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow AC$
$C \rightarrow A$
$A \rightarrow C$

F'
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow A$
$D \rightarrow C$
$C \rightarrow A$
$A \rightarrow C$

F''
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow A$
~~$D \rightarrow C$~~
$C \rightarrow A$
$A \rightarrow C$

F'''
$A \rightarrow B$
$AB \rightarrow D$
$D \rightarrow A$
$D \rightarrow C$
$C \rightarrow A$
$A \rightarrow C$



G
$A \rightarrow B$
$A \rightarrow C$
$A \rightarrow D$
$C \rightarrow A$
$D \rightarrow A$

Source: Garcia-Molina et al. (2007), §3.1.1

Projecting FDs

Projecting Functional Dependencies

Let R be a relation with attributes C and functional dependencies F.
Also, let S be a projection of R onto attributes C'.

The set of functional dependencies, F', of S, can be calculated by:

1. Calculate the closure of every non-empty subset $c \subseteq C'$ using F
2. Initialise F' as $c \rightarrow c^+$ for all c, dropping attributes not in S
3. Simplify the resulting set of $2^{|C|}$-1 FD's into a minimal basis

Source: Garcia-Molina et al. (2007), §3.2.8

Projecting Functional Dependencies

Assume that S(A,B,C) is a projection of R(A,B,C,D,E) and that R had the FD's below. Calculate F'.

$A \rightarrow B$
$B \rightarrow D$
$CD \rightarrow E$

1. Calculate the closure of every non-empty subset $c \subseteq C'$ using F
2. Initialise F' as $c \rightarrow c^+$ for all c, dropping attributes not in S
3. Simplify the resulting set of $2^{|C|}$-1 FD's into a minimal basis

Source: Garcia-Molina et al. (2007), §3.2.8

Projecting Functional Dependencies

Assume that S(A,B,C) is a projection of R(A,B,C,D,E) and that R had the FD's below. Calculate F'.

$A \rightarrow B$
$B \rightarrow D$
$CD \rightarrow E$

$A^+ = \{A,B,D\}$
$B^+ = \{B,D\}$
$C^+ = \{C\}$
$AB^+ = \{A,B,D\}$
$AC^+ = \{A,B,C,D,E\}$
$BC^+ = \{B,C,D\}$

1. Calculate the closure of every non-empty subset $c \subseteq C'$ using F
2. Initialise F' as $c \rightarrow c^+$ for all c, dropping attributes not in S
3. Simplify the resulting set of $2^{|C|}-1$ FD's into a minimal basis

Source: Garcia-Molina et al. (2007), §3.2.8

Projecting Functional Dependencies

Assume that S(A,B,C) is a projection of R(A,B,C,D,E) and that R had the FD's below. Calculate F'.

$A \rightarrow B$
$B \rightarrow D$
$CD \rightarrow E$

$A \rightarrow AB$
$B \rightarrow B$
$C \rightarrow C$
$AB \rightarrow AB$
$AC \rightarrow ABC$
$BC \rightarrow BC$

1. Calculate the closure of every non-empty subset $c \subseteq C'$ using F
2. Initialise F' as $c \rightarrow c^+$ for all c, dropping attributes not in S
3. Simplify the resulting set of $2^{|C|}-1$ FD's into a minimal basis

Source: Garcia-Molina et al. (2007), §3.2.8

Projecting Functional Dependencies

Assume that S(A,B,C) is a projection of R(A,B,C,D,E) and that R had the FD's below. Calculate F'.

$A \rightarrow B$
$B \rightarrow D$
$CD \rightarrow E$

$A \rightarrow B$

1. Calculate the closure of every non-empty subset $c \subseteq C'$ using F
2. Initialise F' as $c \rightarrow c^+$ for all c, dropping attributes not in S
3. Simplify the resulting set of $2^{|C|}-1$ FD's into a minimal basis

Source: Garcia-Molina et al. (2007), §3.2.8

Projecting FD's: Example 2

Let's project R(A,B,C,D,E) onto S(A,C,E) and then project the following FD's onto S as well:

$A \rightarrow B$
$B \rightarrow D$
$CD \rightarrow E$

1. Calculate the closure of every non-empty subset $c \subseteq C'$ using F
2. Initialise F' as $c \rightarrow c^+$ for all c, dropping attributes not in S
3. Simplify the resulting set of $2^{|C|}$-1 FD's into a minimal basis

Source: Garcia-Molina et al. (2007), §3.2.8

Projecting FD's: Example 2

Let's project R(A,B,C,D,E) onto S(A,C,E) and then project the following FD's onto S as well:

$A \rightarrow B$
$B \rightarrow D$
$CD \rightarrow E$

$A^+ = \{A, B, D\}$
$C^+ = \{C\}$
$E^+ = \{E\}$
$AC^+ = \{A,B,C,D,E\}$
$AE^+ = \{A,B,D,E\}$
$CE^+ = \{C,E\}$
$ACE^+ = \{A,B,C,D,E\}$

1. Calculate the closure of every non-empty subset $c \subseteq C'$ using F
2. Initialise F' as $c \rightarrow c^+$ for all c, dropping attributes not in S
3. Simplify the resulting set of $2^{|C|}-1$ FD's into a minimal basis

Source: Garcia-Molina et al. (2007), §3.2.8

Projecting FD's: Example 2

Let's project R(A,B,C,D,E) onto S(A,C,E) and then project the following FD's onto S as well:

$A \rightarrow B$
$B \rightarrow D$
$CD \rightarrow E$

$A^+ = \{A, B, D\}$
$C^+ = \{C\}$
$E^+ = \{E\}$
$AC^+ = \{A,B,C,D,E\}$
$AE^+ = \{A,B,D,E\}$
$CE^+ = \{C,E\}$
$ACE^+ = \{A,B,C,D,E\}$

$A \rightarrow A$
$C \rightarrow C$
$E \rightarrow E$
$AC \rightarrow ACE$
$AE \rightarrow AE$
$CE \rightarrow CE$
$ACE \rightarrow ACE$

1. Calculate the closure of every non-empty subset $c \subseteq C'$ using F
2. Initialise F' as $c \rightarrow c^+$ for all c, dropping attributes not in S
3. Simplify the resulting set of $2^{|C|}-1$ FD's into a minimal basis

Source: Garcia-Molina et al. (2007), §3.2.8

Projecting FD's: Example 2

Let's project R(A,B,C,D,E) onto S(A,C,E) and then project the following FD's onto S as well:

$A \rightarrow B$
$B \rightarrow D$
$CD \rightarrow E$

$A^+ = \{A, B, D\}$
$C^+ = \{C\}$
$E^+ = \{E\}$
$AC^+ = \{A,B,C,D,E\}$
$AE^+ = \{A,B,D,E\}$
$CE^+ = \{C,E\}$
$ACE^+ = \{A,B,C,D,E\}$

$A \rightarrow A$
$C \rightarrow C$
$E \rightarrow E$
$AC \rightarrow ACE$
$AE \rightarrow AE$
$CE \rightarrow CE$
$ACE \rightarrow ACE$

$AC \rightarrow E$

Observe that this was not in our original set!
We would not get this with our "rule of thumb"!

1. Calculate the closure of every non-empty subset $c \subseteq C'$ using F
2. Initialise F' as $c \rightarrow c^+$ for all c, dropping attributes not in S
3. Simplify the resulting set of $2^{|C|}-1$ FD's into a minimal basis

Source: Garcia-Molina et al. (2007), §3.2.8

Project the FDs onto Theatres

MovieShowings(title, theatre, city)
Theatres(theatre,city)

theatre → city
title city → theatre

We will revisit this example in the next lesson (3NF)

Allotted Time: 120 seconds